Project Design Phase-I
Solution Architecture

Date	06-03-2026
Team ID	Srashti Patel
Project Name	Heart Disease Analysis
Maximum Marks	4 Marks

Solution Architecture

The solution architecture for the Heart Disease Data Analysis project focuses on transforming raw healthcare data into meaningful visual insights using Tableau and Tableau Prep Builder. The system collects patient health records, prepares and cleans the data, and presents analytical dashboards that help identify heart disease risk patterns.

The architecture connects data sources, data preparation tools, and visualization platforms to ensure smooth data processing and reporting.

Architecture Overview

1. Data Source Layer

- Contains raw heart disease dataset.
- Includes patient attributes like age, cholesterol, blood pressure, heart rate, etc.
- Data stored in CSV or structured dataset format.

2. Data Preparation Layer (Tableau Prep Builder)

- Cleans and transforms raw dataset.
- Handles missing values and incorrect entries.
- Creates calculated fields and structured tables.
- Outputs a refined dataset for analysis.

3. Data Storage Layer

- Prepared dataset stored as Tableau Extract (.hyper) or structured dataset.
- Ensures fast query processing and efficient visualization.

4. Visualization Layer (Tableau Desktop)

- Connects to prepared dataset.
- Creates dashboards, charts, and analytical views.
- Enables exploration of heart disease risk indicators.

5. Reporting & Insights Layer

- Interactive dashboards show patterns and trends.
- Helps identify high-risk patient groups.
- Provides decision support for healthcare insights.

Data Flow in the System

1. Heart disease dataset is collected from healthcare records or public datasets.
2. The dataset is imported into Tableau Prep Builder for cleaning and transformation.
3. Processed data is exported as a structured dataset or Tableau extract.
4. Tableau Desktop connects to the processed data.
5. Interactive dashboards are created for visualization and analysis.
6. Insights are generated for identifying heart disease patterns and risk factors.

Solution Architecture Diagram:-

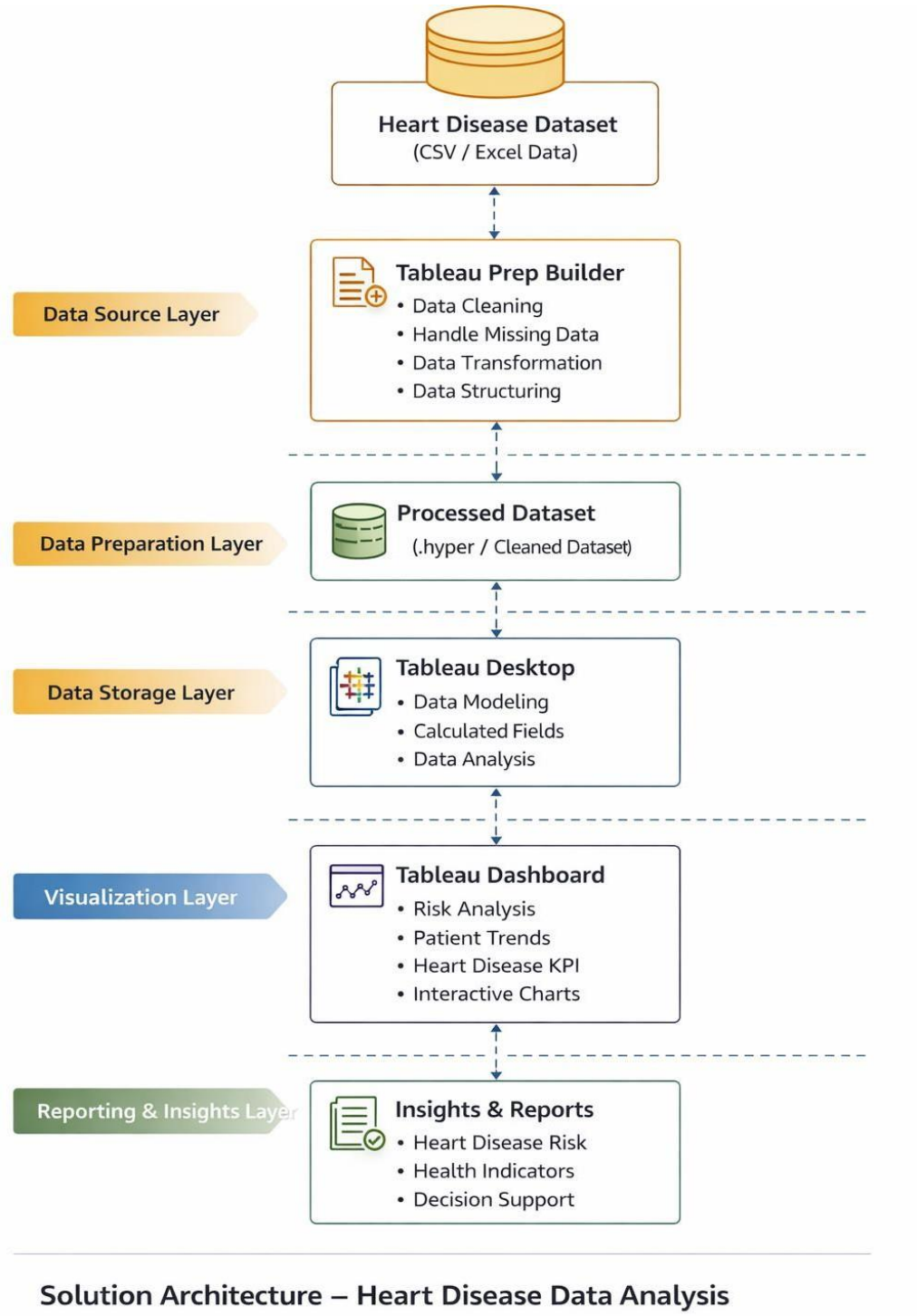

Figure 1: Architecture and data flow of the system

Reference: https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/